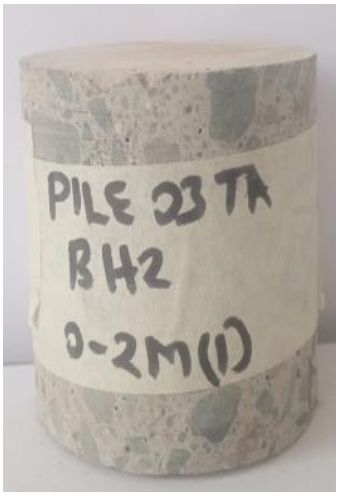
Cransfield Materials Testing Center
Quality Assurance

P.O Box 59401-00200,
Nairobi, Kenya.
Runda Grove, House No. 278
Tel: 0703410973/0733725775
Email: info@cmtc.co.ke
Website: www.cmtc.co.ke

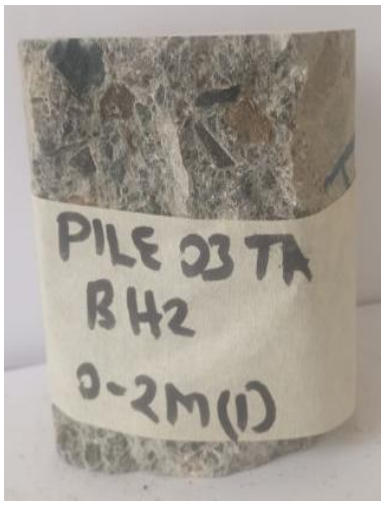
COMPRESSIVE STRENGTH OF CONCRETE CORES
(BS EN 12504-1:2009)

PROJECT NAME: PROPOSED NSSF TWIN TOWERS, NAIROBI COUNTY						
CORE DETAILS						
Sample No.: PILE 03	BH2	**Depth(m):**	0.0-2.0	Date: 9/10/2026	**Class of Concrete (N/mm^2)**	**45 MPA**
Age of Concrete:	over 28 days	**Location:**	TOWER A			
TEST DETAILS						
Core Diameter (mm):	82.0	**Height Before trimming (mm):**	150.0	**Height After trimming (mm):**		82
Core Weight in Air (g):	963.0	**Density (Kg/m3):**	2223	**C(H/D)-correction factor for aspect ratio**		1
Area (m^2):	0.0053	**Volume (m3):**	0.0004	**Cdia-correction factors for core diameter**		1
Maximum Load (KN):	188.7	**Actual Compressive Strength (N/mm2):**	35.7	**Cd- correction factor for core extraction Accuracy**		1.1
Length/ Diameter ratio	1.00	**Cylinder Strength to Cube Strength**	1.25	**Ca-correction factors for the presence of steel:**		1.00
of Core at preparations:		**Conversion Ratio**		**Corrected Insitu Cube Strength (N/mm2):**		49.1

Failure Mode



BEFORE TEST



AFTER TEST

Comments: Sparsely distributed aggregates with maximum size approximately 20mm
Excess Voidage : **1.00%**

CRANSFIELD MATERIAL TESTING P.O.BOX 59401-00100 NAIROBI

Tested by:	PUAL N	**Date reported:**	17-Oct-26	**Checked by:**	

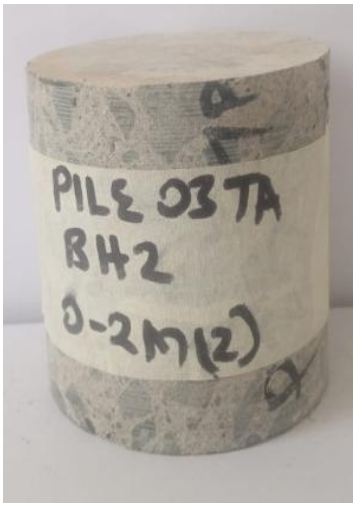
Cransfield Materials Testing Center
Quality Assurance

P.O Box 59401-00200,
Nairobi, Kenya.
Runda Grove, House No. 278
Tel: 0703410973/0733725775
Email: info@cmtc.co.ke
Website: www.cmtc.co.ke

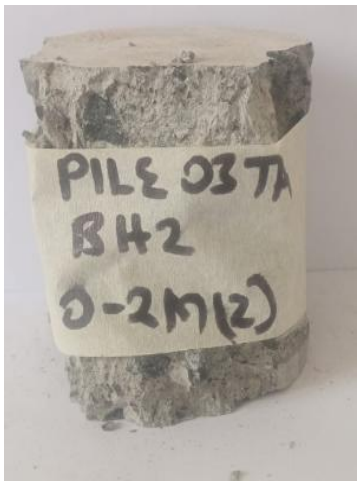
COMPRESSIVE STRENGTH OF CONCRETE CORES
(BS EN 12504-1:2009)

PROJECT NAME: PROPOSED NSSF TWIN TOWERS, NAIROBI COUNTY						
CORE DETAILS						
Sample No.: PILE 03	BH2	**Depth(m):**	0.0-2.0	Date: 9/10/2026	**Class of Concrete (N/mm^2)**	**45 MPA**
Age of Concrete:	over 28 days	**Location:**	TOWER A			
TEST DETAILS						
Core Diameter (mm):	82.0	**Height Before trimming (mm):**	150.0	**Height After trimming (mm):**		82
Core Weight in Air (g):	981.0	**Density (Kg/m3):**	2264	**C(H/D)-correction factor for aspect ratio**		1
Area (m^2):	0.0053	**Volume (m3):**	0.0004	**Cdia-correction factors for core diameter**		1
Maximum Load (KN):	190.8	**Actual Compressive Strength (N/mm2):**	36.1	**Cd- correction factor for core extraction Accuracy**		1.1
Length/ Diameter ratio	1.00	**Cylinder Strength to Cube Strength**	1.25	**Ca-correction factors for the presence of steel:**		1.00
of Core at preparations:		**Conversion Ratio**		**Corrected Insitu Cube Strength (N/mm2):**		49.7

Failure Mode



BEFORE TEST



AFTER TEST

Comments: Sparsely distributed aggregates with maximum size approximately 20mm
Excess Voidage : **1.00%**

Tested by:	PUAL N	**Date reported:**	17-Oct-26	**Checked by:**	

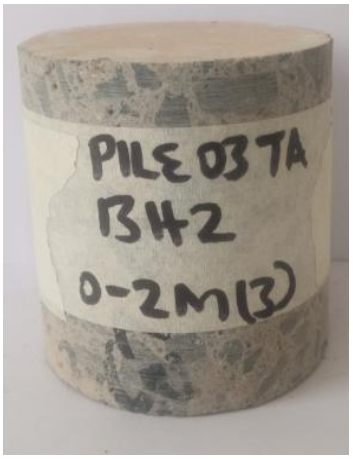
Cransfield Materials Testing Center
Quality Assurance

P.O Box 59401-00200,
Nairobi, Kenya.
Runda Grove, House No. 278
Tel: 0703410973/0733725775
Email: info@cmtc.co.ke
Website: www.cmtc.co.ke

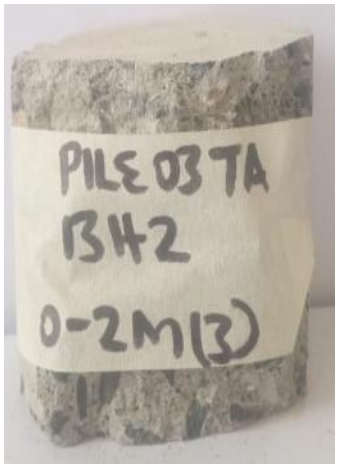
COMPRESSIVE STRENGTH OF CONCRETE CORES
(BS EN 12504-1:2009)

PROJECT NAME: PROPOSED NSSF TWIN TOWERS, NAIROBI COUNTY						
CORE DETAILS						
Sample No.: PILE 03	BH2	**Depth(m):**	0.0-2.0	Date: 9/10/2026	**Class of Concrete (N/mm^2)**	**45 MPA**
Age of Concrete:	over 28 days	**Location:**	TOWER A			
TEST DETAILS						
Core Diameter (mm):	82.0	**Height Before trimming (mm):**	150.0	**Height After trimming (mm):**		82
Core Weight in Air (g):	1026.0	**Density (Kg/m3):**	2368	**C(H/D)-correction factor for aspect ratio**		1
Area (m^2):	0.0053	**Volume (m3):**	0.0004	**Cdia-correction factors for core diameter**		1
Maximum Load (KN):	192.6	**Actual Compressive Strength (N/mm2):**	36.5	**Cd- correction factor for core extraction Accuracy**		1.1
Length/ Diameter ratio	1.00	**Cylinder Strength to Cube Strength**	1.25	**Ca-correction factors for the presence of steel:**		1.00
of Core at preparations:		**Conversion Ratio**		**Corrected Insitu Cube Strength (N/mm2):**		50.1

Failure Mode



BEFORE TEST



AFTER TEST

Comments: Sparsely distributed aggregates with maximum size approximately 20mm
Excess Voidage : **1.00%**

Tested by:	PUAL N	**Date reported:**	17-Oct-26	**Checked by:**	

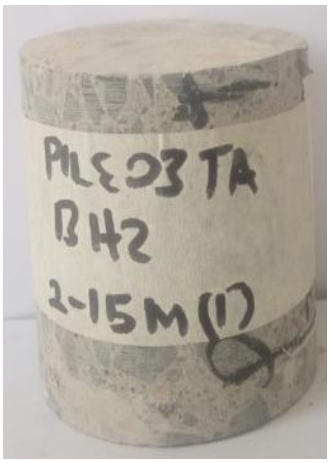
Cransfield Materials Testing Center
Quality Assurance

P.O Box 59401-00200,
Nairobi, Kenya.
Runda Grove, House No. 278
Tel: 0703410973/0733725775
Email: info@cmtc.co.ke
Website: www.cmtc.co.ke

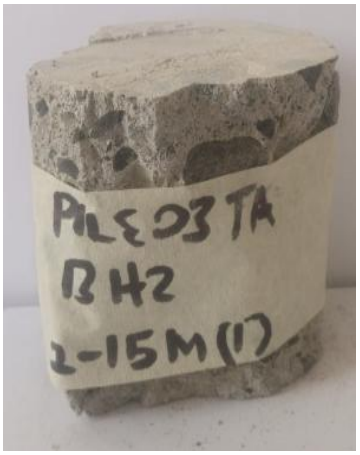
COMPRESSIVE STRENGTH OF CONCRETE CORES
(BS EN 12504-1:2009)

PROJECT NAME: PROPOSED NSSF TWIN TOWERS, NAIROBI COUNTY						
CORE DETAILS						
Sample No.: PILE 03	BH2	**Depth(m):**	2.0-15.0	Date: 9/10/2026	**Class of Concrete (N/mm^2)**	**45 MPA**
Age of Concrete:	over 28 days	**Location:**	TOWER A			
TEST DETAILS						
Core Diameter (mm):	82.0	**Height Before trimming (mm):**	150.0	**Height After trimming (mm):**		82
Core Weight in Air (g):	1019.0	**Density (Kg/m3):**	2352	**C(H/D)-correction factor for aspect ratio**		1
Area (m^2):	0.0053	**Volume (m3):**	0.0004	**Cdia-correction factors for core diameter**		1
Maximum Load (KN):	186.9	**Actual Compressive Strength (N/mm2):**	35.4	**Cd- correction factor for core extraction Accuracy**		1.1
Length/ Diameter ratio	1.00	**Cylinder Strength to Cube Strength**	1.25	**Ca-correction factors for the presence of steel:**		1.00
of Core at preparations:		**Conversion Ratio**		**Corrected Insitu Cube Strength (N/mm2):**		48.6

Failure Mode



BEFORE TEST



AFTER TEST

Comments: Sparsely distributed aggregates with maximum size approximately 20mm
Excess Voidage : **1.00%**

Tested by:	PUAL N	**Date reported:**	17-Oct-26	**Checked by:**	

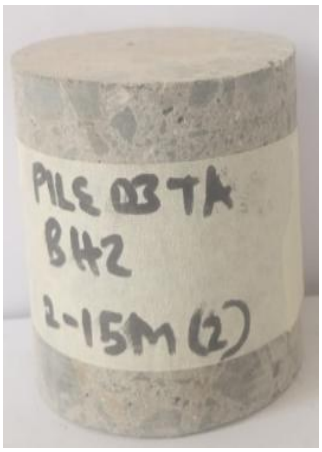
Cransfield Materials Testing Center
Quality Assurance

P.O Box 59401-00200,
Nairobi, Kenya.
Runda Grove, House No. 278
Tel: 0703410973/0733725775
Email: info@cmtc.co.ke
Website: www.cmtc.co.ke

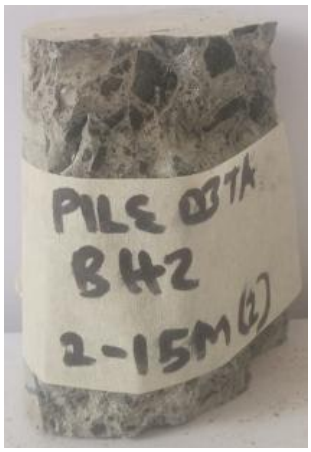
COMPRESSIVE STRENGTH OF CONCRETE CORES
(BS EN 12504-1:2009)

PROJECT NAME: PROPOSED NSSF TWIN TOWERS, NAIROBI COUNTY						
CORE DETAILS						
Sample No.: PILE 03	BH2	**Depth(m):**	2.0-15.0	Date: 9/10/2026	**Class of Concrete (N/mm^2)**	**45 MPA**
Age of Concrete:	over 28 days	**Location:**	TOWER A			
TEST DETAILS						
Core Diameter (mm):	82.0	**Height Before trimming (mm):**	150.0	**Height After trimming (mm):**		82
Core Weight in Air (g):	1037.0	**Density (Kg/m3):**	2394	**C(H/D)-correction factor for aspect ratio**		1
Area (m^2):	0.0053	**Volume (m3):**	0.0004	**Cdia-correction factors for core diameter**		1
Maximum Load (KN):	188.7	**Actual Compressive Strength (N/mm2):**	35.7	**Cd- correction factor for core extraction Accuracy**		1.1
Length/ Diameter ratio	1.00	**Cylinder Strength to Cube Strength**	1.25	**Ca-correction factors for the presence of steel:**		1.00
of Core at preparations:		**Conversion Ratio**		**Corrected Insitu Cube Strength (N/mm2):**		49.1

Failure Mode



BEFORE TEST



AFTER TEST

Comments: Sparsely distributed aggregates with maximum size approximately 20mm
Excess Voidage : **1.00%**

CRANSFIELD MATERIAL TESTING

Tested by:	PUAL N	**Date reported:**	17-Oct-26	**Checked by:**	

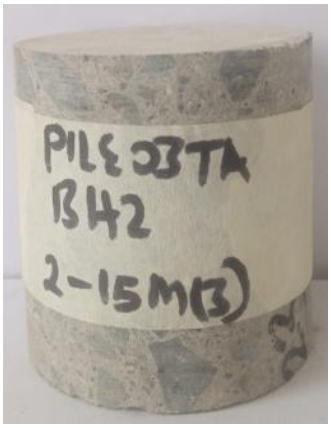
Cransfield Materials Testing Center
Quality Assurance

P.O Box 59401-00200,
Nairobi, Kenya.
Runda Grove, House No. 278
Tel: 0703410973/0733725775
Email: info@cmtc.co.ke
Website: www.cmtc.co.ke

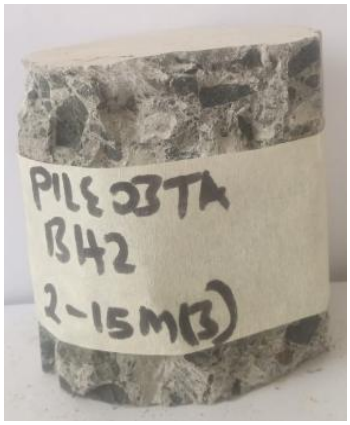
COMPRESSIVE STRENGTH OF CONCRETE CORES
(BS EN 12504-1:2009)

PROJECT NAME: PROPOSED NSSF TWIN TOWERS, NAIROBI COUNTY						
CORE DETAILS						
Sample No.: PILE 03	BH2	**Depth(m):**	2.0-15.0	Date: 9/10/2026	**Class of Concrete (N/mm^2)**	**45 MPA**
Age of Concrete:	over 28 days	**Location:**	TOWER A			
TEST DETAILS						
Core Diameter (mm):	82.0	**Height Before trimming (mm):**	150.0	**Height After trimming (mm):**		82
Core Weight in Air (g):	936.0	**Density (Kg/m3):**	2161	**C(H/D)-correction factor for aspect ratio**		1
Area (m^2):	0.0053	**Volume (m3):**	0.0004	**Cdia-correction factors for core diameter**		1
Maximum Load (KN):	198.0	**Actual Compressive Strength (N/mm2):**	37.5	**Cd- correction factor for core extraction Accuracy**		1.1
Length/ Diameter ratio	1.00	**Cylinder Strength to Cube Strength**	1.25	**Ca-correction factors for the presence of steel:**		1.00
of Core at preparations:		**Conversion Ratio**		**Corrected Insitu Cube Strength (N/mm2):**		51.5

Failure Mode



BEFORE TEST



AFTER TEST

Comments: Sparsely distributed aggregates with maximum size approximately 20mm
Excess Voidage : **1.00%**

CRANSFIELD MATERIAL TESTING
P.O BOX 59401-00100 NAIROBI

Tested by:	PUAL N	**Date reported:**	17-Oct-26	**Checked by:**	

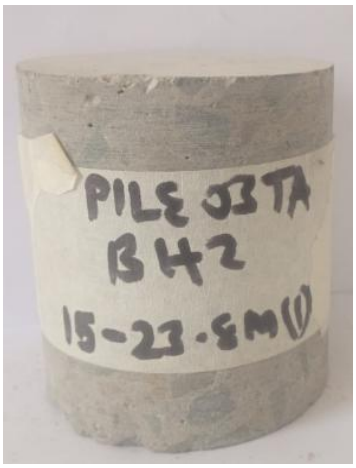
Cransfield Materials Testing Center
Quality Assurance

P.O Box 59401-00200,
Nairobi, Kenya.
Runda Grove, House No. 278
Tel: 0703410973/0733725775
Email: info@cmtc.co.ke
Website: www.cmtc.co.ke

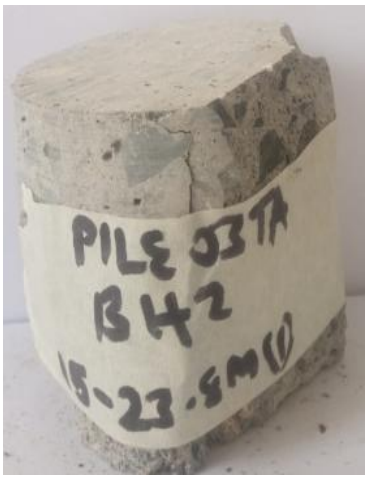
COMPRESSIVE STRENGTH OF CONCRETE CORES
(BS EN 12504-1:2009)

PROJECT NAME: PROPOSED NSSF TWIN TOWERS, NAIROBI COUNTY						
CORE DETAILS						
Sample No.: PILE 03	BH2	**Depth(m):**	15.0-20.6	Date: 9/10/2026	**Class of Concrete (N/mm^2)**	**45 MPA**
Age of Concrete:	over 28 days	**Location:**	TOWER A			
TEST DETAILS						
Core Diameter (mm):	82.0	**Height Before trimming (mm):**	150.0	**Height After trimming (mm):**		82
Core Weight in Air (g):	975.0	**Density (Kg/m3):**	2251	**C(H/D)-correction factor for aspect ratio**		1
Area (m^2):	0.0053	**Volume (m3):**	0.0004	**Cdia-correction factors for core diameter**		1
Maximum Load (KN):	190.9	**Actual Compressive Strength (N/mm2):**	36.1	**Cd- correction factor for core extraction Accuracy**		1.1
Length/ Diameter ratio	1.00	**Cylinder Strength to Cube Strength**	1.25	**Ca-correction factors for the presence of steel:**		1.00
of Core at preparations:		**Conversion Ratio**		**Corrected Insitu Cube Strength (N/mm2):**		49.7

Failure Mode



BEFORE TEST



AFTER TEST

Comments: Sparsely distributed aggregates with maximum size approximately 20mm
Excess Voidage : **1.00%**

CRANSFIELD MATERIAL TESTING

Tested by:	PUAL N	**Date reported:**	17-Oct-26	**Checked by:**	

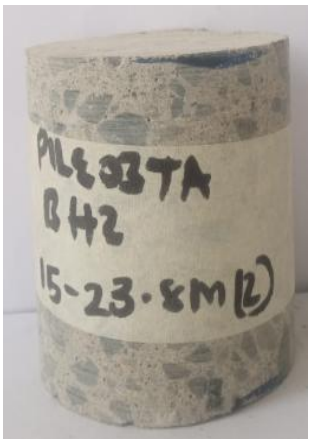
Cransfield Materials Testing Center
Quality Assurance

P.O Box 59401-00200,
Nairobi, Kenya.
Runda Grove, House No. 278
Tel: 0703410973/0733725775
Email: info@cmtc.co.ke
Website: www.cmtc.co.ke

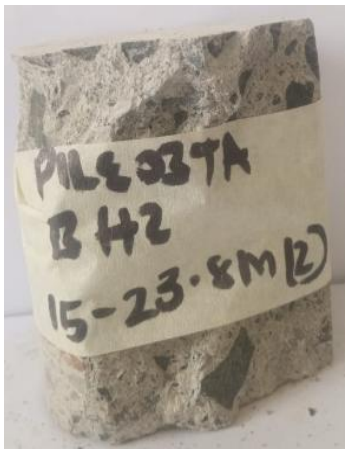
COMPRESSIVE STRENGTH OF CONCRETE CORES
(BS EN 12504-1:2009)

PROJECT NAME: PROPOSED NSSF TWIN TOWERS, NAIROBI COUNTY						
CORE DETAILS						
Sample No.: PILE 03	BH2	**Depth(m):**	15.0-20.6	Date: 9/10/2026	**Class of Concrete (N/mm^2)**	**45 MPA**
Age of Concrete:	over 28 days	**Location:**	TOWER A			
TEST DETAILS						
Core Diameter (mm):	82.0	**Height Before trimming (mm):**	150.0	**Height After trimming (mm):**		82
Core Weight in Air (g):	981.0	**Density (Kg/m3):**	2264	**C(H/D)-correction factor for aspect ratio**		1
Area (m^2):	0.0053	**Volume (m3):**	0.0004	**Cdia-correction factors for core diameter**		1
Maximum Load (KN):	187.6	**Actual Compressive Strength (N/mm2):**	35.5	**Cd- correction factor for core extraction Accuracy**		1.1
Length/ Diameter ratio	1.00	**Cylinder Strength to Cube Strength**	1.25	**Ca-correction factors for the presence of steel:**		1.00
of Core at preparations:		**Conversion Ratio**		**Corrected Insitu Cube Strength (N/mm2):**		48.8

Failure Mode



BEFORE TEST



AFTER TEST

Comments: Sparsely distributed aggregates with maximum size approximately 20mm
Excess Voidage : **1.00%**

CRANSFIELD MATERIAL TESTING
P.O BOX 59401-00100 NAIROBI

Tested by:	PUAL N	**Date reported:**	17-Oct-26	**Checked by:**	

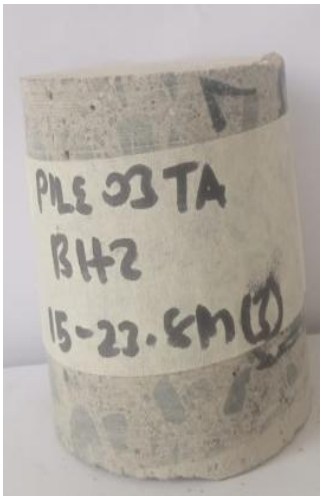
Cransfield Materials Testing Center
Quality Assurance

P.O Box 59401-00200,
Nairobi, Kenya.
Runda Grove, House No. 278
Tel: 0703410973/0733725775
Email: info@cmtc.co.ke
Website: www.cmtc.co.ke

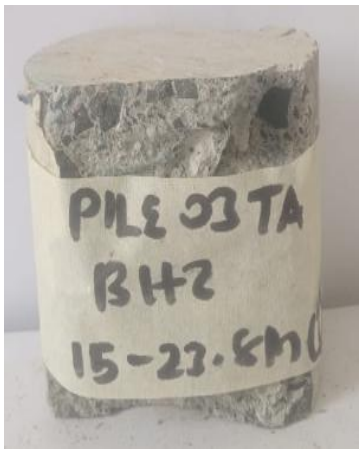
COMPRESSIVE STRENGTH OF CONCRETE CORES
(BS EN 12504-1:2009)

PROJECT NAME: PROPOSED NSSF TWIN TOWERS, NAIROBI COUNTY						
CORE DETAILS						
Sample No.: PILE 03	BH2	**Depth(m):**	15.0-20.6	Date: 9/10/2026	**Class of Concrete (N/mm^2)**	**45 MPA**
Age of Concrete:	over 28 days	**Location:**	TOWER A			
TEST DETAILS						
Core Diameter (mm):	82.0	**Height Before trimming (mm):**	150.0	**Height After trimming (mm):**		82
Core Weight in Air (g):	1023.0	**Density (Kg/m3):**	2361	**C(H/D)-correction factor for aspect ratio**		1
Area (m^2):	0.0053	**Volume (m3):**	0.0004	**Cdia-correction factors for core diameter**		1
Maximum Load (KN):	200.7	**Actual Compressive Strength (N/mm2):**	38.0	**Cd- correction factor for core extraction Accuracy**		1.1
Length/ Diameter ratio	1.00	**Cylinder Strength to Cube Strength**	1.25	**Ca-correction factors for the presence of steel:**		1.00
of Core at preparations:		**Conversion Ratio**		**Corrected Insitu Cube Strength (N/mm2):**		52.2

Failure Mode



BEFORE TEST



AFTER TEST

Comments: Sparsely distributed aggregates with maximum size approximately 20mm
Excess Voidage : **1.00%**

CRANSFIELD MATERIAL TESTING
P.O BOX 59401-00100 NAIROBI

Tested by:	PUAL N	**Date reported:**	17-Oct-26	**Checked by:**	